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Reducing Achievement Gaps in Academic Writing for Latinos and English Learners in Grades 7–12

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Reducing Achievement Gaps in Academic Writing for Latinos and English Learners in Grades
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Abstract

This study reports two years of findings from a randomized controlled trial designed to replicate and demonstrate the efficacy of an existing, successful professional development program, the Pathway Project, that uses a cognitive strategies approach to text-based analytical writing. Building on an earlier randomized field trial in a large, urban, low SES district where 98% of the students were Latino and 88% were mainstreamed English learners (ELs) at the intermediate level of fluency, the project aimed to help secondary school students, specifically Latinos and mainstreamed ELs, in another large, urban, low SES district to develop the academic writing skills called for in the rigorous Common Core State Standards for English Language Arts. The Pathway Project draws on well documented instructional frameworks that support approaches that incorporate strategy instruction to enhance students' academic literacy. Ninety-five teachers in sixteen secondary schools were stratified by school and grade and then randomly assigned to the Pathway or control group. Pathway teachers participated in forty-six hours of training to help students write analytical essays. Difference-in-differences and regression analyses revealed significant effects on student writing outcomes in both years of the intervention (Year 1 $d=0.48$ and Year 2 $d=0.60$). Additionally, Pathway students had higher odds than control students of passing the California High School Exit Exam in both years.

Keywords: Latino Students, English Learners, Teacher Professional Development, Impact Studies, Writing Assessments

In its vision of what it means to be literate in the twenty-first century, the Common Core State Standards for English Language Arts (CCSS-ELA) prioritize the ability to analyze and interpret challenging texts using academic discourse in extended pieces of writing. In addition to specifying specific standards for each grade level, the CCSS-ELA present College and Career Readiness Anchor Standards for Reading and Writing in grades K-5 and grades 6-12 that define the skills and understandings all students must demonstrate. These include the ability to “read closely to determine what the text says explicitly and to make logical inferences from it,” and to “write arguments to support claims in an analysis of substantive topics or texts, using valid reasoning and relevant and sufficient evidence” (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010). As is evident from these anchor standards, the CCSS-ELA set a high bar for all students and place a premium on the ability to analyze and interpret challenging texts and to write about those texts using academic discourse in extended pieces of writing. However, results from the most recent administration of the 2011 National Assessment of Educational Progress in writing (U. S. Department of Education, NCES, 2012a) indicate that today’s secondary students face considerable challenges in meeting these standards. For example, only 27% of all 12th graders and 11% of Hispanic¹ students scored at the level of proficient or above in writing. Most alarming is that only 1% of English Learners (ELs) scored at the level of proficient or above. Given that by 2020, one in four children enrolled in America’s K-12 public schools will be Latino (Maxwell, 2012) and that ELs in grades 7-12 are the fastest growing segment of the K-12 student population (Francis, Rivera, Lesaux, Keiffer, & Rivera, 2006), these disparities are worrisome. Since writing is a gatekeeper for college admission and a “threshold skill” for hiring and promotion for salaried workers (National

¹ Note: We use the term “Hispanic” whenever the data is referenced as such in the primary source (e.g. NAEP data uses the term Hispanic). All other times we use the term “Latino.”

Commission on Writing for America's Families, Schools, and Colleges, 2004), failure to close these achievement gaps in academic writing will have serious social and economic consequences.

Who Are English Learners?

The nation lacks a uniform definition and classification of English learners. Hence, ELs are called many names including English language learners, limited English proficient students, language or linguistic minority students, and second language learners. Forty years ago, many viewed ELs as a relatively homogeneous group with similar instructional needs. This stereotype has endured despite large demographic changes. However, ELs today are a diverse group with unique experiences and backgrounds (Harklau, Losey, & Siegal, 1999; Matsuda, Ortmeir-Hooper & You, 2006). Currently, ELs constitute approximately 10% of the total K-12 population (U.S. Department of Education, NCES, 2012b). Although ELs in the United States speak more than 350 languages, 73% speak Spanish as their first language (Batalova & McHugh, 2010), 40% have origins in Mexico (Hernandez, Denton, & Macartney, 2008), and 60% of ELs in Grades 6 through 12 come from low-income families (Batalova, Fix, & Murray, 2005; Capps et al., 2005). At the same time that EL enrollments have increased in U.S. public schools, researchers and policymakers have highlighted large literacy gaps based on students' English language proficiency.

The largest numbers of ELs in our schools today are referred to as Long-Term English Learners or LTELs (Menken & Kleyn, 2009). According to Olsen (2010), these are students who have been educated in the United States since age six, are doing poorly in school, and have major gaps in knowledge because their schooling was disrupted. In Olsen's study of 175,734 ELs, the majority (59%) were LTELs who were failing to acquire academic language and struggling to do

well in high school. They may come from homes where the primary language is not English, but they themselves may speak English only or they may switch between multiple languages and still have features in their writing attesting to their multilingual status (Valdés, 2001). Limited in their knowledge of academic registers in any language, these students are often mainstreamed into regular English language arts classrooms. Given the many demands that academic writing places on all students and the few opportunities to practice, ELs need explicit instruction and ongoing support as they strive to become college and career ready. As Goldenberg (2013) reminds us, “It should be clear that despite progress in understanding how to improve teaching and learning for the millions of ELs in our schools, many gaps remain. The challenges posed by the Common Core State Standards make those gaps more glaring” (p. 10).

Overview of the Pathway Project Professional Development Program

The treatment is an intensive 46-hour professional development program in which secondary teachers learn how to integrate cognitive strategy instruction into process writing to improve students’, specifically Latinos’ and mainstreamed ELs’, interpretive reading and text-based analytical writing by: 1) using a cognitive strategies approach to reading and writing instruction; 2) instructing students to revise a pre-test on-demand writing assessment into multiple draft essays; and 3) receiving ongoing support from experienced Pathway teachers who serve as coaches to teachers in the experimental condition. Results from earlier studies (Olson & Land, 2007; Olson, Land, Anselmi, & Aubuchon, 2010; Kim et al., 2011; Olson et al., 2012) suggest that integrating strategy instruction within a text-based approach to analytical writing can enhance Latinos’ and ELs’ writing ability.

The efficacy of the Pathway Project owes much to the teachers-teaching-teachers model of the National Writing Project with its inherent respect for the capacity of practitioners to

generate and use knowledge to inform and improve their practice. Ongoing and sustained professional development based on the analysis of student work contributes to the establishment of a professional learning community dedicated to the academic progress of students. This collaboration creates a network of language arts classes in which highly trained teachers prepare all students, but specifically Latinos and ELs, to develop the interpretive reading and analytical writing abilities necessary for academic success.

Why Take a Cognitive Strategies Approach to Teaching Text-Based Analytical Writing to Latinos and Mainstreamed English Learners?

The Pathway Project professional development takes a cognitive strategies approach to reducing the achievement gap between Latinos, ELs, and their native English-speaking peers in the area of text-based academic writing. Numerous reports from policy centers and blue-ribbon panels “implicate poor understandings of cognitive strategies as the primary reason why adolescents struggle with reading and writing” (Conley, 2008, p. 84; Graham, 2006; Snow & Biancarosa, 2003). Further, research conducted over the past fifteen years on the content of college courses and instructor expectations indicates that cognitive strategy use is the key to college and career readiness (Conley, 2013). The cognitive strategies intervention that is the focus of this study is grounded in a wide body of research on what experienced readers and writers do when they construct meaning from and with texts. Countless studies demonstrate the efficacy of cognitive strategy use in reading (Block & Pressley, 2002; Duke & Pearson, 2002; National Institute of Child Health and Human Development, 2000; Paris, Wasik, & Turner, 1991; Tierney & Pearson, 1983; Tierney & Shanahan, 1991). Similarly, Graham and Perin (2007) indicate that strategy instruction is the most effective of eleven key elements of writing instruction ($d=.82$) for all students and particularly for students who find writing challenging.

Increasingly, recent instructional frameworks and recommendations also support approaches that incorporate strategy instruction to advance ELs' development of English (Calderon, Slavin, & Sanchez, 2011; Francis et al., 2006; Goldenberg, 2008; Schleppegrell, 2009). Short and Fitzsimmons (2007) hypothesize that strategy instruction is especially effective for ELs because it provides them with an explicit focus on language, increases their exposure to academic texts, makes the texts they read comprehensible, gives them multiple opportunities to affirm or correct their understanding and use of language, assists them in retrieving new language features and in using these features for academic purposes, and provides them with the means of learning language on their own, outside of class. They further hypothesize that adolescent ELs of an intermediate level of English proficiency, who represent the majority of Long Term English Learners (LTELs) in California (Olsen, 2010) have sufficient proficiency to benefit from strategy instruction (Echevarria, Short, & Vogt, 2008; Short & Fitzsimmons, 2007) because they possess the language proficiency required to use the types of cognitive strategies that will provide them access to the higher order cognitive reading and writing tasks encountered in regular content instruction. Explicitly teaching strategic reading and writing behaviors to ELs can help them engage with complex texts and convey those interpretations in well-reasoned essays to meet the CCSS-ELA (August & Shanahan, 2006; Bunch, Kibler, & Pimentel, 2012; Francis et al., 2006; Goldenberg, 2008).

Although research-based practices for developing cognitive strategies are recommended as the “treatment for literacy reform in middle and high schools” (Conley, 2008, pp. 84-85), very little of this type of instruction occurs in school, especially for Latinos in low-SES schools and for ELs (Block & Pressley, 2002; Conley, 2013; Graham & Perin, 2007; Kong & Pearson, 2003; Vaughn & Klinger, 2004). Indeed, researchers have noted a “growing inequality” in classroom

instruction where students designated as “honors students” are exposed to rigorous academic work designed to promote higher literacy, whereas lower achievers, children of the poor, and second language learners often receive instruction that places a premium on the “transmission of information, providing very little room for the exploration of ideas, which is necessary for the development of deeper understanding” (Applebee et al., 2003, p. 689). According to a Carnegie Corporation report, inadequate educator capacity and the limited use of research-based instructional practices prevent ELs from learning academic English at a level necessary to meet content standards in English language arts (Short & Fitzsimmons, 2007). This negatively impacts their ability to participate fully in educational programs, successfully complete coursework, and achieve the academic outcomes of which they are capable.

Theoretical Framework and Theory of Change

The Pathway Project is informed by cognitive, sociocognitive and sociocultural theory. In their cognitive process theory of writing, Flower and Hayes (1981) posit that writing is best understood “as a set of distinct thinking processes which writers orchestrate and organize during the act of composing” (p. 275), including: planning, organizing, goal setting, translating, monitoring, reviewing, evaluating, and revising. They liken these processes to a “writer’s tool kit” (p. 285) which is not constrained by any fixed order or series of stages. Similarly, Tierney and Pearson (1983) propose that “reading and writing are essentially similar processes of meaning construction” (p. 568) and that readers compose meaning, creating drafts of their understanding, just like writers do. The characteristics they identify as being essential to the meaning construction of readers are planning, drafting, aligning, revising, and monitoring. Concurring with Flower and Hayes, they note that this cognitive process involves continuous, recurring, and recursive transactions between readers and writers. Shanahan (in press) notes that

this shared cognition model conceptualizes reading and writing as being built upon a similar foundation and that both readers and writers utilize procedural knowledge about how to access, use, and generate information intentionally and enact cognitive strategies such as predicting, questioning, and summarizing in the act of meaning construction.

However, writers and readers do not construct meaning in a vacuum. Langer (1991) asserts that literacy comprises more than the individual's ability to read and write. Drawing on the work of Vygotsky (1986), she suggests that literacy is the ability to think and reason like a literate person *within a particular society*. In other words, literacy is culture-specific and meaning is socially constructed. She writes,

Within social settings, both at home and in school students learn how literacy is used and how literate knowledge is communicated—what counts as a literacy event and what literacy behaviors ‘look like,’ what literacy related values are respected, and what literacy habits are to be cultivated. As children learn to engage in literate behaviors to serve the functions and reach the ends they see modeled around them, they become literate in a culturally specific way; they use certain cognitive strategies to structure their thoughts and complete their tasks, and not others (p.17).

From a sociocognitive perspective, teachers should pay more attention to the social purposes to which literacy skills are applied and should go beyond delivering lessons on content to impart strategies for thinking necessary to complete literacy tasks first with guidance and, ultimately, independently.

Finally, sociocultural theory views meaning as being “negotiated at the intersection of individuals, culture, and activity” (Englert, Mariage, & Dunsmore, 2006). Three tenets of sociocultural theory are applicable to the intervention:

1) Cognitive Apprenticeships:

Sociocultural theory promotes the power of cognitive apprenticeships in which novices learn literate behaviors through the repeated modeling of more mature, experienced adults or peers who make the tacit knowledge of meaning construction visible through explicit instruction, as well as think alouds to provide access to strategies and tools demonstrated by successful readers and writers (Vygotsky, 1986). Jerome Bruner (1978) used the term scaffolding to describe this “tutorial assistance” provided to the apprentice by the veteran.

2) Procedural Facilitators and Tools:

A second tenet of sociocultural theory is that teachers are most effective when they lead cognitive development in advance of what students can accomplish alone by presenting challenging material along with procedural and facilitative tools to help readers and writers address those cognitive challenges. These mental, linguistic and physical tools help bridge the gap between what a student can perform alone and what Vygotsky refers to as the zone of proximal development (1986) and can include: notational systems, think sheets, graphic organizers, prompts, planning strategies, mnemonics, text structures, cue cards, and more.

3) Community of Practice:

Third, and perhaps most importantly, sociocultural theory values the establishment of communities of practice in which teachers actively encourage students to collaborate and

provide ongoing opportunities and thoughtful activities that invite students to engage in shared inquiry. Tompkins (2010) compares this classroom community approach to the difference between owning and renting a home:

In a classroom community, students and teachers are joint ‘owners’ of the classroom. Students assume responsibility for their own learning and behavior, work collaboratively with classmates, complete assignments, and care for the classroom. In traditional classrooms, in contrast, the classroom is the teacher’s and the students are simply renters for the school year (p. 16)

(Adapted from Englert, Mariage, and Dunsmore, 2006)

In the Pathway Project, all three theoretical frames—cognitive, sociocognitive, and sociocultural—apply to working with teachers as well as to working with students as teachers are introduced to the cognitive strategies that underlie the reading and writing process, provided with a variety of tools to make strategy use “visible” to students, and engage in professional learning community school teams to implement what they learn during professional development in their classrooms.

The theory of change underlying the Pathway Project intervention is that providing teachers with sustained, high-quality professional development focused on helping students improve their academic writing would prompt them to implement cognitive strategies instruction in their classrooms (proximal outcome), resulting in enhanced student performance on a Pathway-developed pre/post, timed writing assessment (intermediate outcome), transferring to improvement on summative assessments of state content standards (distal outcome) (See Figure 1).

[Insert Figure 1 here]

This logic model was developed based on findings from an earlier four-year randomized controlled trial conducted in the Santa Ana Unified School District (SAUSD), a large, urban, low-SES school district with 98% Latino students, 88% mainstreamed ELs at the intermediate level of fluency or above, and only 7% of English-Only (EO) speakers. This prior study yielded effect sizes of .35 in Year 1 and .67 in Year 2 on an on-demand assessment of academic writing. It also showed that the Pathway Project had a small but statistically significant impact on the writing subtest of the California Standards Test (CST) (Kim et al., 2011; Olson et al., 2012). However, because 98% of the students in SAUSD were Latino and only 7% were classified as English Only, it was not possible to compare the progress of Latinos or ELs with other subgroups. Like SAUSD, the Anaheim Union High School District (AUHSD)—the site of the current study—is a large, low-SES, urban district serving over 33,000 students, but the student population is more diverse (3% African American, 16% Asian, 64% Hispanic, 12% White, 1% Pacific Islander, 3% Other; 23% English Only), allowing for comparisons between race/ethnicity and language subgroups, which was not possible in the prior (SAUSD) study.

Our present study is designed to answer the following research questions:

1. To what extent will teachers' participation in the Pathway Project professional development program improve academic outcomes for secondary students in grades 7-12 on an on-demand writing assessment and a state-mandated high school exit exam in English language arts?
2. To what extent does the intervention have a differential effect on academic outcomes for targeted race/ethnicity and language proficiency subgroups on the measures noted above?

Given the considerable cognitive burden that text-based academic writing places on students in general and ELs in particular, we hypothesized that taking a cognitive strategies approach to literacy instruction would result in a positive impact on student writing outcomes.

Method

Study Design

This study utilized a multi-site, cluster, randomized field trial where secondary schools were the sites, teachers were randomly assigned to Pathway or control in the intervention, and one of each teacher's classes was randomly selected to participate in the study. Students were randomly assigned to each teacher's classes.

Participants

In the 2012-2013 school year, AUHSD enrolled 33,000 students in Grades 7 to 12, 60% of whom were Latino, 66% of whom were English Learners (ELs) or Reclassified Fluent English Proficient (RFEPs), and 66% of whom were eligible for Free or Reduced-Price Lunch (FRPLs). AUHSD classifies its ELs based on an initial language placement exam using the California English Language Development Test (CELDT). A reclassified fluent English proficient (RFEP) student is a student that was initially identified as needing English learner support and subsequently scoring at the 50th percentile or below on the CELDT, but through a combination of receiving proficient scores on the CELDT (scoring 75th percentile or above), strong academic performance in courses, proficient scores on other standardized tests, and by parent recommendations the student is placed into mainstreamed English classrooms.

Teacher participants

In the summer of 2012, 100 teachers from 16 secondary schools in the district (8 middle and 8 high schools) were recruited to participate in the study. Of these, 95 consented to

participate in the study and were randomly assigned, separately for each grade within each school, to the Pathway or control condition. The study sample consisted of 49 Pathway teachers and 46 control teachers. To avoid crossover effects, Pathway teachers were instructed not to share program materials with control teachers. Observation of control classrooms showed no evidence that any sharing had occurred. On average, teachers in the study had 14.82 years of total teaching experience and 77% percent had earned a master's degree. Although teachers received their baccalaureate degrees from 21 different undergraduate institutions, a majority (56%) graduated from a California State University. There was no statistically significant difference between Pathway teachers and control teachers in the total years of teaching experience ($p = .45$), the percentage who graduated with a bachelor's degree from a California State University ($p = .79$), or the percentage who earned a master's degree ($p = .15$). These findings indicate that Pathway and comparison group teachers were similar on observed teacher characteristics measured at baseline.

Student participants

In the first step of a three step process, school counselors employed a software program to randomly assign students to teachers' classrooms. Next, recruited teachers were randomly assigned to either Pathway or control conditions. Finally, in the third step, due to resource constraints, for both groups of teachers one class was selected as the focal class for the study. For both Pathway and control teachers, the class selected was the one that had the greatest percentage of English Learners, and where the students had the English language proficiency necessary to write in English and thereby profit from the enhanced training received by the teachers who participated in the program.

Comparison of Pathway and Control Classrooms at Baseline

Table 1 shows descriptive statistics for Pathway and control group students in the first year of the project, as well as a test of the significance of between-group differences. The groups have a similar distribution across grades, with a slightly smaller percentage of 7th graders in the Pathway group. The gender distribution is well matched. Latinos and Asians constitute the great majority of all students in the schools, and the percentages of these students in each group are closely matched. However, there is a small tendency for a higher percentage of Whites in the control group and a higher percentage of Other Race students in the Pathway group. English Learners are a slightly higher percentage of the control group (21% versus 18%) and Initially Fluent English Proficient students are a slightly higher percentage of the Pathway group (9% versus 6%). The two groups have the same percentage (71%) of students qualified for free or reduced price lunch. Most important, there are no significant between-group differences in pre-test scores on the Academic Writing Assessment (AWA) and the California Standards Test for English Language Arts (CST-ELA). Overall, these statistics show that the random assignment for the first year of the project was successful in producing Pathway and control groups that were well balanced on student characteristics.

[Insert Table 1 here]

The Pathway and control groups were not as well matched at the beginning of the second year of the project. This is because the design called for both Pathway and control teachers to continue in their respective roles. The goal was to see whether Pathway teachers would be able to achieve even larger positive effects following a year of experience with the program. However, there was teacher attrition between the years (Pathway teachers declined from 49 to 41 and controls from 46 to 40), and this altered the demographics of their students (where the Pathway students were receiving the program for the first time and the controls had never

experienced the program). This change in the demographic match of Pathway and control classrooms is unfortunate, but was impossible to avoid because schools in the study had both differential demographics and differential teacher attrition. The results are shown in Table 2. The Pathway students in the second year of the project were more concentrated in 9th and 12th grades; the control students in 8th and 11th grade. The Pathway group had slightly more female students than the control group. The Pathway group also had significantly fewer Latino and more Asian students than the control group; it also had a somewhat higher share of IFEP students. Finally, the Pathway group had a significantly lower share of free and reduced price lunch students than the controls (66% versus 72%). They also had significantly higher AWA test scores, 5.48 versus 5.11, a 7.2 percent difference. These are all variables that will be controlled in our regression analysis estimates of program effects.

[Insert Table 2 here]

Year 1 analysis sample for student achievement analysis.

The analysis sample was smaller than the baseline sample because there were two Pathway teachers who did not administer the Academic Writing Assessment (AWA) at post-test. However, there were no statistically significant differences between the students of these two teachers compared to the rest of the sample in terms gender, free/reduced lunch eligibility, language proficiency status, and the number of Latino students. As a result, the final sample size for the AWA was 93 teachers (in grades 7-12) in 47 Pathway classrooms and 46 control classrooms. To adhere to budget constraints, we scored pre- and post-test on-demand writing assessments for a random sample of 20 students within each class. There was no significant difference on pre-test AWA writing scores for the Pathway ($M = 5.72$, $SD = 2.05$) and control classrooms ($M = 5.69$, $SD = 1.92$, $p = .73$).

Year 2 analysis sample for student achievement analysis on post-test measures.

The Year 2 analysis sample in 2013-2014 ($n=81$ teachers) was smaller than the final Year 1 sample in 2012-2013 ($n=93$) due to teacher attrition. As a result, the final sample for the Academic Writing Assessment (AWA) in Year 2 consisted of 40 control classrooms and 41 Pathway classrooms. Although 14% of the teacher sample was lost to attrition from Year 1 to Year 2, the attrition rate did not differ significantly between the two groups ($p<0.42$). To adhere to budget constraints, we scored post-test on-demand essays for a random sample of 20 students within each class.

Description of the Pathway Project Professional Development Program

Pathway teachers in the intervention participated in 46 hours of training each school year (via six 6-hour released days interspersed throughout the school year and five 2-hour afterschool sessions) focused on methods for helping Latinos and mainstreamed ELs to develop the academic literacy necessary to meet the CCSS-ELA, with special emphasis on interpretive reading and analytical writing. These included: literary response and analysis, comprehension and analysis of informational nonfiction texts, and development of clear, coherent, focused essays. Training was led by the developers of the Pathway Project and supported by literacy coaches who participated as Pathway teachers in a previous quasi-experimental research study (Olson & Land, 2007). There are three core components of the Pathway Project: (1) training in the use of cognitive strategies tool kit and curriculum materials, (2) Pathway Project activities focused on the revision of the pre-test on-demand writing assessment into a multiple draft essay, and (3) coaching from a more experienced, veteran teacher previously trained in the program on how to integrate a cognitive strategies approach into the existing English language arts curriculum.

Cognitive strategies tool kit and curriculum materials.

Strategy instruction in this intervention occurred within the context of teaching reading and writing as a process and involves pre-reading, during reading, and post-reading activities as well as prewriting, planning, drafting, sharing, revising, and editing activities. The Pathway Project was initially developed over an eight-year period in SAUSD (Olson & Land, 2007), in response to the specific needs of teachers serving a rapidly increasing population of ELs. At the time, few interventions were available that addressed the writing challenges of ELs. For example, in a recent meta-analysis of writing instruction (Graham, McKeown, Kiuvara, & Harris, 2012), only three studies (out of 115) for which an effect size could be computed included ELs. Therefore, the Pathway Project and its year-long, six full-day professional development schedule constantly evolved as new research became available. During the first two PD days, teachers were introduced to a model of the cognitive strategies that make up a reader's and writer's mental tool kit in Figure 2. These thinking tools or acts of mind directly map on to the CCSS-ELA Anchor Standards for College and Career Readiness in Reading and Writing that call for students to be able to do the following as they read and write about complex texts: summarize, make inferences, analyze, interpret, draw conclusions, evaluate, assess, revise, and reflect. The CCSS-ELA leaves to curriculum developers, states, and teachers' "professional judgment and experience" the determination of what "tools and knowledge" students will need in order to meet these challenging new standards (National Governors Association Center for Best Practices & Council of Chief State School Officers, 2010, p. 4). The Pathway Project focused on these tools. Teachers used the following analogy to introduce the concept of the tool kit:

When we read, we have thinking-tools or cognitive strategies inside our heads that we access to construct meaning. Researchers say that when we read, we're

composing, just as when we write. What they mean is that while we read, we're creating our own draft of the story in our heads and as we keep reading and come across something we didn't expect to happen or suddenly make a big discovery about what something means, we start on a second draft of our understanding. So, when you think of yourself as a reader or writer, think of yourself as a craftsman, skilled in making things with your hands, but instead of reaching into a metal tool kit for a hammer or a screwdriver to construct or build tangible or real objects you can actually see, you're reaching into your mental tool kit to construct meaning from or with words.

[Insert Figure 2 here]

To reinforce the tool kit analogy, teachers received wall posters with visuals representing the cognitive strategies, and students received bookmarks as well as 8½" x 11" copies of cognitive strategies sentence starters that illustrate what goes on in the mind of a reader or writer in the act of meaning construction. For example, a sentence starter for revising meaning is "At first I thought—but now I..." and a starter for reflecting and relating is, "So, the big idea is..." To build students' declarative knowledge of what cognitive strategies are, teachers presented scaffolded lessons called "tutorials" (Bruner, 1978) in which they introduced each of the tools in the tool kit to students within the context of reading and writing about high-interest literary or nonfiction texts. To enhance their procedural knowledge of how to implement the strategies, students received instruction on how to make marginal annotations to interpret complex texts, and kept reading logs with key quotes from the texts they were reading and commentary on those quotes. Finally, to foster conditional knowledge of when to use a cognitive strategy, which strategy to use, and why, students were taught to think aloud in response to complex texts while a partner

recorded their responses and then labeled their strategy use, as well as wrote metacognitive reflections describing the cognitive strategies they used in order to form interpretations about texts and write analytical essays. The Pathway Project provided a wide array of teacher-tested and easy-to-use paper and computer-based materials as models of curriculum and instruction. Because these materials were designed to be presented to students across the grade levels (7-12) and with varying degrees of language proficiency, teachers were given time to meet in grade-level groups and as school teams to discuss how to modify the materials to meet their specific students' needs. Additionally, optional scaffolded materials were provided for struggling students. For example, a unit entitled the symbolism detour was designed for ELs who had difficulty understanding figurative language. Teachers were encouraged to gradually withdraw the instructional scaffolding embedded in the materials as students demonstrated their ability to implement cognitive strategies in reading and writing more independently (Pearson & Gallagher, 1983).

Formative assessment and revision of pre-test

Second, teachers learned how to use results from the on-demand analytical writing pre-test (the Analytical Writing Assessment—AWA) to provide instruction in text-based analytical writing. To that end, the professional development focused on preparing students to read, make inferences, and form interpretations about complex literary and nonfiction texts and to convey interpretations in thoughtful, well-organized essays that present a clear thesis supported with appropriate textual evidence. The centerpiece of the Pathway Project is an extensive set of materials shared during Days 3 and 4 of the professional development program, focused on the revision of the students' pre-test writing assessment (a text-based analytical essay) into a multiple draft essay. Student performance on this timed, on-demand pre-test essay was used to

inform the Pathway Project as teachers engaged in analyzing students' work and identifying students' strengths and areas for growth (Biancarosa & Snow, 2004; Black & Wiliam, 1998).

Two of the ten Common Core College and Career Readiness Standards for Reading for grades 6-12 focus on determining and analyzing themes in both single and multiple texts. Additionally, in a national survey of 2,351 high school and college teachers administered by The College Board to identify skills teachers thought were most important for students entering college, respondents rated identifying the theme of a text and making inferences and drawing conclusions as the most important reading skills, and writing a clear, coherent essay using supporting details as the most important writing skill students should possess (Milewski, Johnsen, Glazer, & Kubota, 2005). Despite its importance, many students have an inadequate grasp of what a theme is. For this reason, the prompts for the Year 1 AWA (2012-2013) focused on analyzing theme in two works of fiction, the short stories, "The Medicine Bag," by Virginia Driving Hawk Sneve (1991) and "The Scarlet Ibis" by James Hurst (1960), one at pre-test and one at post-test. Order effects were controlled by counterbalancing the two timed writing assessments across classrooms each year. In Year 2, teachers received additional training in using a cognitive strategies approach to teaching informational texts and literary nonfiction, genres prioritized in the CCSS-ELA which students may have had less experience studying and writing about in their English language arts classes. The Year 2 AWA prompts required students to analyze two newspaper articles, "Sometimes, the Earth is Cruel" by Leonard Pitts (2010) and "The Man in the Water" by Roger Rosenblatt (1982), develop a theme statement about the author's message, and evaluate the author's purpose for writing each article.

Pre-test essays were used for both formative and summative purposes. Based on the teachers' analysis of students' pre-test essays, lessons were implemented to address students'

needs relative to the CCSS-ELA. For example, paper 1C124 in Appendix A, an exploration of theme in “The Medicine Bag” written by an 8th grade, Latina, re-designated EL student in October, contains a number of areas for improvement that are typical of the pre-tests we received. First, the paper is written as one, long paragraph without a formal introduction, main body and conclusion. Graham & Perin (2007) note that “teaching adolescents strategies for planning, revising, and editing their compositions has shown a dramatic effect on the quality of students’ writing (p. 16). During the revision activities, students were taught a planning strategy, HoT S-C Team (see Figure 3), to lead into their essays with a hook – a quote, question, description, or statement to make people think, followed by a TAG (title, author and genre), a statement summarizing the conflict, and a thesis containing their theme statement.

[Insert Figure 3 here]

While this strategy might be perceived as too formulaic for advanced writers, novice essay writers need to be exposed to form-making before engaging in form-breaking. Students were also introduced to transition words to develop coherence and use as a bridge to their main body and conclusion and practiced inserting appropriate transition words into cloze exercises to become familiar with the use and position of words like *in addition*, *however*, *nevertheless*, *whereas*, *on the other hand*, etc.

Another very common shortcoming in pre-test essays was the over-reliance on summarizing. As has been widely reported, ELs who have been in English Language Development programs often receive instruction that focuses primarily on literal comprehension. Consequently, they tend to rely on retelling when writing a text-based analytical essay as a way to prove that they understood what they read rather than offering interpretation and commentary to support their argument. As Scardamalia and Bereiter (1987) point out, novice, inexperienced

and struggling writers use a simplified version of the idea-generation process they call knowledge-telling which consists of retrieving information from long term memory and converting the writing task into simply regurgitating what is known about a topic. More experienced writers, on the other hand, engage in a complex composing process known as knowledge-transformation in which they analyze the writing task and plan what to say and how to say it in accordance with rhetorical, communicative and pragmatic constraints. Noting that students whose text production system corresponds to the knowledge-telling model “need more than encouragement to revise” (p. 156), Scardamalia and Bereiter (1987) suggest that they need modeling, lessons in planning and writing for an audience, and “insight into their own composting processes” (p. 165). One way to help students move from knowledge-telling to knowledge-transformation is to help them make their thinking visible after they have composed a first draft of an essay using a color-coding process.

Teachers first designated three colors for the types of assertions that comprise a text-based analytical essay and said the following:

Plot summary reiterates what is obvious and known in a text. Reiterate means to repeat in order to make something very clear. Plot summary is yellow because it’s like the sun. It makes things as plain as day. We need some plot summary to orient our reader to the facts, but we don’t need to retell the entire story.

Commentary is blue like the ocean because the writer goes beneath the surface of things to look at the deeper meaning to offer opinions, interpretations, insights, and “Ah-Ha’s.” Supporting detail is green because like the color, it brings together the facts of the text (yellow) with your interpretation of it (blue). It is

what glues together plot summary and commentary. It's your evidence to support your claims, including quotations from the text.

The next step was to model the process of color-coding. Since half the students in the study wrote to one prompt and half to the other at pre-test, we selected a third text as a training text. In Year 1, "The Horned Toad" by Gerald Haslam (2000) was used as the training text. The sample paragraph about "The Horned Toad" (below) illustrates how one codes for summary, supporting detail, and commentary:

When the narrator's horned toad was crushed on the pavement, Grandma consoles him over the loss of her pet. (Yellow)

She joins him in grieving and strokes his back and then she picks up the horned toad and mutters, "The poor little beast." (Green)

Like the horned toad, Grandma is also out of place. (Blue)

The horned toad is symbolic of Grandma because they are kindred spirits who need to return to where they belong. (Blue)

After students were introduced to the color-coding system, they practiced coding sample essays on theme in "The Horned Toad" that were marginal/not pass (1-3 on a 6 point scale) and adequate to strong pass (4-6 on a 6 point scale). Starting with the weaker paper, students noticed that most of the sentences fell into the yellow category whereas the stronger paper had a balance of yellow, green and blue. Students then applied the color-coding strategy to their own first drafts to visibly see whether they had simply summarized or whether they had provided ample textual evidence and commentary. The coded draft then became a visible guide for revision.

At pre-test, a majority of the students in the study had difficulty distinguishing between a topic and a theme and tended to use topic words like *death*, *loss*, or *selfishness* to identify the

theme in “The Scarlet Ibis” or *family, relationships, or tradition*, as the student in paper 1C124 does, to describe theme in “The Medicine Bag,” rather than to present a well developed theme statement, or they failed to discuss the theme altogether. To help students to distinguish between a topic and a theme, teachers first provided the following explanation:

A story’s theme is different from its topic or subject. The topic is simply what it’s about. The theme is the author’s point about a topic. Think of a topic as the What of the story and the theme as the So what? To identify a theme, sometimes it helps to brainstorm a list of topics or big ideas in a story. Common topics for themes that you’ll find in stories are usually abstract nouns that deal with human relationships and include terms like belonging, courage, family, friendship, hope, identity, prejudice, respect, revenge, and trust. A theme statement must be a complete sentence that states the author’s message about life or about human relationships, such as “Loss can bring people together.” A good theme statement applies to people in general, not just to the specific characters in the story.

To help students practice developing theme statements, teachers showed several short video clips from familiar movies featuring teenage protagonists in situations where a theme could be derived. For instance, in the Disney/Pixar animated film *Brave*, Princess Merida defies an age old custom to be betrothed to her future husband through an arranged marriage orchestrated by her parents. In the clip, during an archery contest in which various suitors compete to win her hand in marriage, Merida grabs the bow and shoots with better precision than any of the would-be grooms. Students generated the following topics and theme statements in response to the clip.

Topics.

Tradition, heritage, culture, chivalry, defiance, marriage, authority, disobedience

Theme statements.

- Sometimes it's necessary to challenge the traditions of one's heritage.
- Control of your life is in your own hands.
- Women shouldn't be considered second class citizens and should have the freedom to choose the person they want to marry.

Thinking of the topic as the “What” and the theme as the “So What?” and practicing with the video clips seemed to turn on a light bulb for students and helped them to create a more definitive theme statement to support with textual evidence.

Another area in which students needed support during the revision process was the use of academic language. Academic language is a key component of effective instruction for ELs (Gersten, et. al., 2007; Goldenberg, 2013; Rivera et al., 2010; Short & Fitzsimmons, 2007) and the explicit instruction of academic English for all students figures prominently in the CCSS-ELA. In analyzing students' pre-tests, we found that, by and large, they used an informal and conversational register of English rather than the more formal, academic register called for in an analytical essay. In other words, they used a “think it/say it” type of “unretouched” and “underprocessed” expression of thought written by a writer to him/herself for him/herself diagnosed by Flower (1979) as writer-based prose rather than a more deliberate and mature structure and style adapted to a reader, or reader-based prose (p. 19). These more informal linguistic features include general words like *nice*, *story*, or *man* instead of more academic or technical ones like *compassionate*, *narrative*, or *novelist*, slang like *stuff*, *guy*, and *blown away*, inappropriate hedges like *kind of* and *sort of* (as the author of paper 1C124 does when she says that Martin was “kinda embarresed” [sic] of his grandfather), informal expressions and markers of spoken English like *ya know* and *by the way*, needless repetition, misspellings, grammatical

errors, contractions, poorly linked sentences with an absence of clear referents, the unnecessary use of I (e.g., *I think*, *I believe*, and *In my opinion* when it is clear that the student is offering an opinion), and simple rather than complex sentences with little sentence variety.

To help students become more conscious of and adept at writing in a register of English suited to a formal audience, we seamed together a “typical” pre-test essay written in informal English and then rewrote the same paper using more formal academic language. Students were instructed to work in pairs to create a chart noting the features of each essay that made it informal or formal. Next, the partners were given the informal passage below and instructed to revise all of the underlined words (which also included some common editing errors) to make them sound more academic.

An informal passage to improve.

The boy in the story really didn't like his Great-Grandma when they first met. She barked orders at him and everyone else. She said scary things about his horned toad, and his mother made him take it back were he found it. Grandma was sorta mean I think. Like she called him el malcriado, the spoiled boy. It seems that the boy kinda started to like his Great-Grandma when she gave him money for candy and for sure when she helped him bury the horned toad. Maybe he would of liked her in the beginning if she had talked English to him. Cause she didn't try in the beginning, he didn't either. A theme for this story might be something like “Never judge a book by its cover.” This means wait til you get to know someone before you make a judgment about whether you like or dislike them.

Appendix B includes the post-test on-demand essay written by student 1C124 in May, after the revision training. Note that her paper: opens with a hook, followed by a TAG (title, author,

genre), and a theme statement; has a clear introduction, main body, and conclusion; includes a blend of summary, evidence, and commentary; uses transition words and academic words such as *narrator* and *quotes*, as well as a sentence variety strategy we taught called adjectives out of order (Noden, 2011). Although the paper still contains some errors in correctness, the growth in the student's academic writing is substantial.

Coaching.

The third core component of the Pathway Project involves coaching. Throughout Year 1, teachers received ongoing support from a retired veteran teacher who had previous exposure to the project during an earlier 8-year quasi-experimental study in SAUSD (Olson & Land, 2007). These coaches conducted three informal, non-evaluative classroom observations and provided teachers with detailed written feedback identifying areas of strength, areas for improvement, and specific suggestions of classroom practices teachers could implement. During Year 2, a lead English language arts teacher from each school who had been groomed during Year 1 of the training assumed the coaching role. Pathway Project coaches attended professional development trainings along with the school team to whom they were assigned and assisted teachers in integrating interpretive reading and analytical writing instruction using the cognitive strategies approach into the lessons in their textbook. Research indicates that when coaching is combined with professional development, teachers are more likely to implement innovations in their classroom (Olson & Land, 2008; Buly, Coskie, Robinson, & Egawa, 2006; Joyce & Showers, 2002).

Description of Business-as-Usual Professional Development Activities.

In contrast, control teachers conducted business as usual, using the district English language arts textbook and core novels for teaching. Both groups attended one full day of

professional development led by district curriculum specialists on protocols for reviewing district benchmark assessments. In Year 2, in addition to district benchmarks, text complexity was also addressed by a district curriculum specialist.

Data Sources and Collection

Data collection took place from September 2012 until June of 2014. Data sources included existing school district records of student demographic information (gender, race/ethnicity, free and reduced-price lunch, and English learner status), and outcome measures including pre- and post-test on-demand writing assessments (AWA) administered by the participant teachers and the California High School Exit Exam (CAHSEE).

Measures and Procedures

Academic Writing Assessment (AWA).

The AWA is a measure developed for the Pathway Project to assess the analytical writing skills of secondary school students. In October of each year, students in the classes of Pathway and control teachers wrote a timed, on-demand, text-based academic essay interpreting a theme in one of two literary selections (“The Scarlet Ibis” and “The Medicine Bag” in Year 1) and one of two nonfiction newspaper articles (“Sometimes, the Earth is Cruel” and “The Man in the Water” in Year 2) to determine their growth as academic writers over time. As mentioned previously, order effects were controlled by counterbalancing the two timed-writing assessments across classrooms in each year. Essays were organized by classrooms and then randomly assigned to trained raters. Each rater scored essays holistically on a 6-point scale to assess the quality and depth of the interpretation, the clarity of the thesis, the organization of ideas, the appropriateness and adequacy of the evidence, sentence variety, and the correct use of English language conventions. Our rubric for scoring the AWA was based on those used to evaluate the

essay portion of the California High School Exit Examination (CAHSEE) (California Department of Education, 2008a), the California STAR 7 Direct Writing Assessment (California Department of Education, 2008b), and the NAEP (ACT, 2007). Each paper received two ratings, yielding a combined score ranging from 2 (1+1) to 12 (6+6) with all papers in which raters disagreed by more than 1 point scored by a third rater. In total there were 60 raters. Each essay was independently scored by two raters, with 52% exact agreement and 94% within one point agreement in Year 1, and 47% exact agreement and 93% within one point agreement in Year 2. Discrepancies of more than one point were resolved by taking the average score of the first two raters and then summing this number with the third rater's score. For example, if the first two raters assigned a score of 2 and 4, and the third rater assigned a score of 3, the final score was 6. Approximately 6% of the papers were scored by a third rater..

California High School Exit Exam (CAHSEE).

In anticipation of the Smarter Balanced Assessment, California elected not to administer the California Standards Test (CST) in 2014. However, the California High School Exit Exam (CAHSEE) was administered to all 10th graders during 2013 and 2014. Tenth graders first sit for this exam during March of each year. The CAHSEE is a standardized literacy assessment that is aligned with the California English/Language Arts academic content standards through grade ten and is a graduation requirement for all high school students. The test consists of multiple choice questions and a timed writing task. The reading portion includes vocabulary, reading comprehension, and analysis of information and literary texts. The writing portion covers writing strategies, applications, and conventions. The writing task is randomly assigned at each test sitting and calls for students to write a response to literature, a biography, an expository piece, or

a persuasive essay. CAHSEE scale scores range from 275-450. A scale score of 350 or higher is needed to pass the CAHSEE.

Fidelity of implementation measure.

To assess the fidelity of implementation, we used the Pathway Project Quality Checklist, an instrument specifically aligned with our intervention. This measure relies on the research base on effective literacy instruction, including principles that appear to enhance student learning across various models and curricula. In their study of primary grade reading instruction, Foorman, Schatschneider, Eakin, Moats, and Francis (2006) have noted wide variance in the quality with which teachers implement curricula but also found that scores on a Likert rating form were linked to literacy outcomes. The teacher-focused measure includes a 6-point Likert scale (1 = little or weak evidence to 6 = impressive evidence) on four items that the observer uses to rate the degree to which the teacher: (1) demonstrates knowledge of subject matter, (2) delivers a lesson that is appropriate for the needs of students, (3) engages students, and (4) uses language arts strategies that are consistent with the Pathway Project. The student-focused measure includes the same 6-point Likert scale on two items that the observer uses to rate the degree to which the students: (1) exhibit command of reading and writing strategies, and (2) are on task and engaged in the lesson.

The fidelity of implementation measure was part of a more extensive form that we used to guide teacher observations during both years of the intervention. During Year 1, we observed 91 of the participating teachers (Pathway and control) for one class period (46 minutes) at two time points (winter 2012 and spring 2013). During Year 2, 74 of the participating teachers (all of whom were also observed in Year 1) were observed following the same procedure as Year 1. We used the same three trained raters both years who were blinded to teachers' status as Pathway

or control. We established inter-observer reliability on our instrument by having pairs of raters rate 20% of the classrooms on all six Likert ratings each year. We found that across the six Likert ratings we had perfect agreement in 86% of the sets of observations. Only 14% of the Likert ratings had any disagreement and they were in at most one of the six observed Likert ratings in an observation session. We take this as strong evidence that we have a reliable, low inference observation tool. Any discrepancies were resolved through discussion between observers.

To determine whether there was observable growth in the use of Pathway Project strategies among teachers who underwent the training, we compared the observation data of Pathway and control teachers who were observed at both time points (winter and spring) during both years of the program (Year 1 and Year 2). In our Year 1 study (2012-2013), there was not a significant difference between control teachers ($M=3.94$, $SD=1.21$) and Pathway teachers ($M=4.39$, $SD=1.02$) in their use of language arts strategies and activities consistent with the intervention at the first observation conducted in winter 2012 ($p=0.10$, see Table 3). On the final observation, there was again not a significant difference ($p=0.18$) in the time spent by Pathway teachers and control teachers in their implementation of Pathway Project-specific strategies and activities. At the first observation in Year 2, there was not a significant difference ($p=0.29$) between control teachers and Pathway teachers. At the final observation, however, Pathway teachers ($M=5.19$, $SD=0.67$) implemented Pathway Project-specific strategies and activities at a significantly higher rate than the control teachers ($M=4.29$, $SD=1.30$, $p<0.01$). A similar pattern occurred when observers scored teachers on the extent to which their students demonstrate command of and effective use of strategies. During the Year 1 winter and spring observations, and during the Year 2 winter observation, there were no significant differences between Pathway

and control teachers' students' scores (see Table 3). By the Year 2 spring observation, there was a significant difference between the two groups ($p < 0.05$). The growth that occurred over time for the Pathway teachers' students' on both measures suggests that teachers needed prolonged exposure to and practice with the strategies in order to fully implement them.

Data Analytic Strategy

Separate calculations were undertaken to estimate program effects for the first and second implementation years in AUHSD. To measure the impact of the Pathway Project on student writing achievement as measured by the Academic Writing Assessment (AWA), we undertook two sets of calculations. The first compared the pre-test to post-test gains of Pathway and control groups for the total analysis sample and for demographic subgroups. The second used the following gain score regression analysis model:

$$(1) Y_2 - Y_1 = \alpha + \beta_1 \text{Pathway} + \beta_2 \text{Controls} + \epsilon$$

In this model, Y_2 represents the student's post-test essay score, Y_1 represents the student's pre-test essay score, Pathway is a dummy variable for placement in the Pathway or control group, and controls include gender, race, grade, language proficiency status, and free/reduced lunch eligibility. The models also included tests for interaction between the Pathway variable and both the ethnic/race and language subgroups. Unstandardized coefficients were used for each of the numeric variables, and the program effect size (Cohen's d) was calculated by dividing the Pathway coefficient by the pooled standard deviation of the pre-test. The model was estimated in STATA 13 using the cluster command with the commands regress and logistic (Rogers 1993; Williams 2000). This methodology adjusts the estimated standard errors using the Huber-White "sandwich" estimator (Huber 1967; White 1980) to correct the standard errors for clustering of students into the classes of Pathway and control teachers. It has been shown that when there are

more than 20 clusters (as is the case in this study), this technique produces similar standard errors to those estimated from a multilevel model (Arceneaux & Nickerson, 2009).

The California High School Exit Exam provided another opportunity to measure the impact of the Pathway Project on teachers and students. The results for the students were reported by the district. Students were coded as passing (1), failing (0), or did not yet take (2). For analyses in each year, the 10th grade students' performances on the CAHSEE were analyzed. We did not analyze the 11th or 12th grade students because the spring of 10th grade is when students first attempt to take the CAHSEE. The numbers of students retaking the test in 11th and 12th grade are much smaller and are subject to selection bias. As with the AWA analyses, we first compared the outcomes for Pathway and control students overall and within each of the demographic subgroups. Following this, we ran logistic regression analyses predicting passing the CAHSEE using models which included the group of control variables and tested for interactions between the Pathway and both the race/ethnicity and language groups. As with the AWA regressions, the models were estimated in STATA, and the Huber-White "sandwich" estimator was used to adjust for the clustering of students into teachers' classrooms. The estimated model is

$$(2) \quad \log(p/1-p) = \alpha + \beta_1 \text{Pathway} + \beta_2 \text{Controls}$$

Results

Results for the Academic Writing Assessment (AWA)

Table 4 shows Year 1 pre- and post-test scores on the AWA for Pathway and control students. These are measured on the AWA scale (from 2-12). Pathway students gained 0.99 points more than control students, which was highly statistically significant. Significant effects were attained for all grade levels except 12th. Program effects were relatively similar for males

and females, although slightly larger for females. Across race/ethnic groups the largest positive effects are for Hispanics and Blacks (1.16 and 1.19, respectively). Those for Whites and Asians, while also statistically significant, are about half those magnitudes (.58 and .51, respectively). Program effects are positive and significant for all the language groups, with the very largest occurring for English Learners. At a magnitude of 1.53, this effect equals the pre-test standard deviation for EL control students. These large gains by ELs suggest that Pathway Project instructional strategies may be particularly beneficial for students still in the process of learning English. The program also produced significant gains for both free and reduced price lunch eligible students, and those who were not eligible. The effects were larger for the lower income (FRPL eligible) students.

[Insert Table 4 here]

Table 5 repeats these calculations for the second year of the program. (To be clear, Pathway students in this sample are in their first year experiencing the program, but Pathway teachers are in their second year with the program. Control students and teachers have never experienced the program.) Once again there is a large, positive, significant effect of the program for the full sample. As during the first year, positive and significant effects occur for males and females, and for Whites, Hispanics, and Asians. As before, the largest gains are for Hispanics, and at 1.47 they are larger than the first year gains for this group. Significant positive effects are found for all four language groups with the largest occurring for RFEPS ($d=0.82$, $p<.000$). Also as before, significant effects are observed both for students who are and are not eligible for free or reduced price lunch, and the larger effects occur for those who are eligible for FRPL.

[Insert Table 5 here]

Table 6 shows these program effects after regression adjustment for the control variables. For each program year the first model shows the Pathway variable alone, the second adds the control variables to the equation, and the third and fourth models add Pathway interactions with race/ethnicity and language groups, respectively. The second models for each year show that, with control variables in the equation, the Pathway effects continue to be positive, significant, and relatively unchanged in magnitude from the first models. For the first implementation year the effect is 0.96 on the AWA scale, which equals an effect size of .48. For the second year, the effect is 1.30 on the AWA scale, which equals an effect size of .60.

The third models show that with these controls, significant, positive program effects are found for Whites in Year 2 (the base category for race/ethnicity), with a larger, marginally significant program effect for Hispanics compared to Whites in both Year 1 and Year 2. This is one of the most important findings in the study; the Pathway Project not only produces significant writing gains for White students, but it produces larger effects for Hispanics. The final model for each year shows that the program produced significant writing gains for English Only students in both program years, as well as larger (and significantly different from English Only) effects for RFEPs in the second year of the program.

[Insert Table 6 here]

Results for the California High School Exit Exam (CAHSEE)

The CAHSEE is first administered to California students in tenth grade, and to avoid selection bias involving students who do not pass it this first time and take it again at a later grade, we restrict the analysis to Pathway and control students in the 10th grade. Table 7 shows these results for the first year of the program; Table 8 shows results for the second year.

Table 7 shows that in the first program year, 75.2% of controls and 83.7% of Pathway students passed the test, for an overall significant, positive program effect of 8.5 percentage points. This effect is large and significant for males [but not for females], for Hispanics, and for English Only students. (We choose to ignore the finding for Other Race students since their sample size totals only 19 students.) The program effect is also large and significant for FRPL, but not for non-FRPL students. These first year results, positive program effects for Hispanics, English Only, and FRPL students, are consistent with the first year results for AWA outcomes. [Insert Table 7 here]

Table 8 repeats these analyses for the second year of the program. This shows a very large effect on the CAHSEE. Fully 87.7% of Pathway 10th graders passed the exam, compared to only 69.3% of the controls, yielding a program effect of 18.4 percentage points, more than twice that achieved in Year 1. Positive program effects are observed for most of the population subgroups, although small sample sizes prevent many of them from achieving statistical significance.

[Insert Table 8 here]

Table 9 uses multivariate logistic regression to adjust for control variables (Models 2 and B) and also test for interactions (models 3, 4, C, and D). The table shows odds ratio coefficients, with t-statistics in parentheses. We also adjusted the standard errors for clustering using the Huber-White “sandwich” estimator. Unfortunately, the result is that the estimated program effects and many of their interactions with race and language now fail to achieve statistical significance. However, the direction and magnitude of the effects are supportive of the findings from Tables 6 and 7. We conclude that the evidence suggests, but does not definitely

demonstrate, that the Pathway Project, in addition to raising scores on the AWA, also increased CAHSEE passing rates, particularly for Hispanics and Asians.

[Insert Table 9 here]

Figures 4 and 5 compare the Year 1 and 2 CAHSEE passing rates for Pathway and control groups with the statewide rates for Hispanics and English learners. Particularly striking is that in both program years both Hispanic and EL Pathway students exceeded the statewide pass rate for these student groups. This shows that the program has been particularly successful with these groups. It is also noteworthy that 71% of AUHSD 10th graders in our sample who took the CAHSEE receive Free and Reduced Priced Lunch whereas only 59% of 10th graders who took the CAHSEE statewide are designated as FRPL, suggesting that the program remains effective for students from a low socio-economic background.

[Insert Figure 4 here]

[Insert Figure 5 here]

Discussion

By 2020, one in four children enrolled in America's K-12 public schools will be Latino (Maxwell, 2012). Currently, Latino students already represent the largest ethnic group in public schools (53.25%) in California (California Department of Education, 2015). At both the national and state level, educational outcomes for Latino students lag behind those of most other ethnic and racial groups, particularly their White peers. Furthermore, 9.2% of the U.S. student population are ELs (U.S. Department of Education, NCES, 2015). California's 1.6 million EL students attending K-12 public schools represent 25% of the state's student population and one-third of all ELs in the country (Payan & Nettles, 2008). But many other states such as Arizona, Texas, Florida, New York, and Illinois serve large and growing numbers of ELs. As is evident

from NAEP data, these students are struggling to achieve proficiency in writing. Given the demographic trends and academic performance of Latinos in general, and ELs in particular, literacy practitioners and researchers are looking for interventions to improve academic outcomes for Latinos and ELs in secondary schools.

In the conclusion of their meta-analysis of writing instruction, Graham and Perin (2007) point out that there is a “serious gap” in the research literature pertaining to secondary adolescents from low-income families, inner-city settings, and/or with low English language proficiency. In addition, Fitzgerald and Amendum (2007) report no empirical studies of 6-12 writing instruction in their meta-analysis that involved 1988-2003 research studies of K-12 writing instruction for ELs in the United States. Given the dearth of research regarding effective literacy instruction for Latinos and mainstreamed ELs in secondary school, we believe our work has the potential to contribute to the scientific knowledge-base regarding strategies that may reduce the achievement gap in writing between Latinos, ELs, and their peers in grades 7-12. Our findings highlight the efficacy of implementing a cognitive strategies approach to reading and writing instruction that makes visible to students the thinking tools accessed by experienced readers and writers during the process of meaning construction. Additionally, our study results are consistent with Taylor et al.’s (2005) findings on the influence of teachers’ practices that encourage cognitive engagement at the elementary level and confirm Langer’s (2000) findings at the secondary level. In particular, our results indicate that teachers can learn to engage Latinos and mainstreamed English Learners in higher-level interpretive reading and analytical writing about texts through direct strategy instruction, modeling of strategy use, and creating opportunities for students to practice and apply these skills through teacher coaching and feedback.

Our main finding is that our randomized controlled trial of the Pathway Project in AUHSD largely replicated the earlier RCT in SAUSD (Year 1 SAUSD $d=.35$, Year 2 SAUSD $d=.67$; Year 1 AUHSD $d=.48$, Year 2 $d=.60$). Further, the demographic diversity of AUHSD enabled us to test whether explicitly training teachers to teach their students strategies to read and write about complex texts, including planning and goal setting, tapping prior knowledge, making connections, forming interpretations, monitoring, revising meaning, reflecting and relating, and evaluating could help to close the achievement gap in writing for Latinos in grades 7-12 as well as improve educational outcomes for ELs. Our impact estimates suggest large, positive effects on the AWA, which is the on-demand writing assessment most closely tied to the Pathway Project, in both Year 1 and Year 2 of the program, as well as positive effects on the CAHSEE pass rates in both years of the program. However, due to particularly small sample sizes for the regression analyses of the CAHSEE, the Pathway effect failed to attain statistical significance in these regressions. We thus consider the CAHSEE effects to be suggestive rather than definitive. To place our results in a broader research context, it is useful to compare the magnitude of the Pathway effect to results from Graham and Perin's 2007 comprehensive meta-analysis of writing instruction for adolescent students. The magnitude of the Pathway effect ($d=.48$ in Year 1 to $d=.60$ in Year 2) is higher than the mean weighted effect size for process writing ($d=.32$) and similar to other writing interventions that include professional development for teachers ($d=.46$). However, the magnitude of the Pathway effect on writing ($d=.48$ in Year 1 to $d=.60$ in Year 2) is smaller than the mean weighted effect size for strategy instruction ($d=.93$). It is unclear precisely why the Pathway effect is smaller, on average, than the effect size for strategy instruction. The complex nature of language development and its effect on strategy use and writing development require further exploration. Students' ability to benefit from strategy

instruction might vary as a function of English proficiency. Additionally, several of the strategy interventions included in the meta-analysis were delivered by research staff, whereas the current intervention was delivered by classroom teachers, and this might have affected the differential outcome.

The AWA means that we report (see Tables 3 and 4) show that while White, Hispanic, and Asian Pathway students all had statistically significant gains from pre-test to post-test, Pathway Hispanics demonstrated larger gains from pre-test to post-test and higher average post-test scores than their White peers in the control group, and they were less than a quarter point behind their White Pathway peers in Year 1. In Year 2, Pathway White and Hispanic students again had statistically significant gains from pre-test to post-test. Pathway Hispanics again showed the largest gains. Further, they had higher average post-test scores than White control students and were on average just half a point (on the AWA) behind their White Pathway peers. These results indicate a substantial reduction in the achievement gap between Latino students and their White peers. Regression analyses confirm the significant gains made by Hispanic students; after controlling for other demographic variables, there was still a marginally significant interaction for AWA gains made by Hispanic students in Pathway in Year 1 and Year 2 (see Table 6).

Similarly, while English learners (EL), Re-designated Fluent English Proficient learners (RFEP), and English Only (EO) students all gained significantly from pre-test to post-test on the AWA, ELs had the greatest pre/post gains of any group in Year 1 and made greater gains than their English Only peers in Year 2. Further, their post-test scores were one-quarter of a point behind their EO control peers, again demonstrating a further closing of the achievement gap (see Tables 3 and 4). Notably, in Year 2, there was a significant, positive interaction for RFEP

students in Pathway compared to English-only speakers (our reference group), which highlights the efficacy of Pathway Project strategies for students whose native language may not be English. Our findings for the CAHSEE generally support these results (Tables 6 and 7), but small sample sizes in the regressions (Table 9) suggest that these results should be taken as suggestive rather than definitive.

What might account for the increase in the AWA effect size from Year 1 to Year 2 and the higher CAHSEE pass rates of Pathway students compared to control students between Year 1 and Year 2? Pressley (2002) has observed that it takes time for teachers to internalize and implement cognitive strategies interventions with confidence and competence. At the end of Year 2, teachers were asked to describe which strategies they felt were most effective in increasing the literacy skills (both reading and writing) of their students. The most highly reported “effective strategy” was the color coding of essays—both students’ own papers and model or anchor papers (24 teachers reported). The next most effective strategies were using the cognitive-strategies tool kit and teaching the introductory and revision tutorials (21 teachers reported each of these as effective). Based on teacher reports such as these and classroom observations that showed notable differences in Pathway and control teachers’ instruction at the end of Year 2, we speculate that teachers were not only more comfortable with implementing the Pathway Project in Year 2, but their recognition of the benefit that their students received in Year 1 motivated them to integrate Pathway strategies into their curriculum with greater fidelity and intensity.

Limitations and Next Steps

Our study had several limitations. One is that, because of differential teacher attrition across schools, which was correlated with differences in the student demographics of these

schools, demographic imbalances between Pathway and control group students occurred in the second year of the study. (However, these demographics were controlled in regression analyses.) A second limitation is that despite the fact that AUHSD is more ethnically and linguistically diverse than SAUSD, only 12% of the students are White and only 23% are EO. Our pre/post writing assessment of students in grades 7-12 had a sample size that was large enough to make comparisons among race/ethnicity and language subgroups. However, in our 10th grade sample of the CAHSEE pass rates, some numbers dropped below a level that was stable enough to make such comparisons. In a current study involving over 8,000 students from four large, urban school districts in the Southern California region, we hope to compare the impact of the Pathway Project on an even more diverse student population. Our research indicates that cognitive strategies instruction has a salutary effect on Latinos and ELs as evidenced by their improved AWA scores and CAHSEE pass rates. However, other factors, including academic vocabulary, grammar, and a range of discourse features, may also merit instruction in English Language arts classes. In addition, many learner characteristics might also affect students' development and use of strategies, including the learners' first and second language proficiency, background knowledge, experience, previous education, and gender. Further research is needed to address the limitations of the current study and to replicate these promising findings in other school districts.

Conclusion

Throughout the nation, literacy practitioners and researchers are looking for effective practices to help Latinos and ELs in secondary school improve their academic writing. The robust findings from this study yield promising results that appear to close the achievement gap in writing for Latinos and help ELs to gain ground by narrowing the achievement gap between ELs and their native English speaking peers. Further, these findings highlight the importance of

sustained ongoing professional development for teachers if they are to effectively teach academic reading and writing to Latinos and ELs.

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Table 1 *Baseline comparison of Pathway and Control Groups, Year 1*

Variable	Control ⁿ¹		Pathway ⁿ²		z	p
	M or %	SD	M or %	SD		
7 th Grade (%)	0.18	0.39	0.16	0.37	-1.84	0.07 [†]
8 th Grade (%)	0.20	0.40	0.21	0.40	0.21	0.83
9 th Grade (%)	0.15	0.35	0.15	0.36	0.57	0.57
10 th Grade (%)	0.20	0.40	0.20	0.40	-0.02	0.98
11 th Grade (%)	0.15	0.36	0.16	0.37	0.83	0.40
12 th Grade (%)	0.12	0.32	0.12	0.33	0.35	0.73
Male(%)	0.51	0.50	0.50	0.50	-0.77	0.44
Hispanic(%)	0.68	0.47	0.68	0.47	0.37	0.71
White (%)	0.13	0.33	0.10	0.30	-2.18	0.03 [*]
Asian (%)	0.17	0.37	0.18	0.38	0.76	0.45
Other (%)	0.05	0.21	0.06	0.24	1.84	0.07 [†]
ELL (%)	0.21	0.41	0.18	0.39	-2.07	0.04 [*]
RFEP (%)	0.44	0.50	0.46	0.50	1.34	0.18
IFEP (%)	0.06	0.24	0.09	0.28	2.89	0.004 ^{**}
EO (%)	0.29	0.46	0.27	0.45	-1.33	0.18
FRPL (%)	0.71	0.45	0.71	0.45	-0.17	0.87
AWA Pre	5.69	1.92	5.72	2.05	-0.34	0.73
CA Standards Test (ELA)	358	55.6	360	56.3	-0.51	0.68

N1=46 teachers, 1493 students

N2=49 teachers, 1705 students

Note: Other=Blacks and Native Americans; Abbreviations are as follows: ELL (English Language Learner), RFEP (Redesignated as Fluent English Proficient), IFEP (Initially Fluent English Proficient), EO (English Only), FRPL (eligible for free/reduced price lunch)

†p<.10, *p<.05, **p<.01, ***p<.001

Table 2 *Baseline comparison of Pathway and Control Groups, Year 2*

Variable	Control ⁿ¹		Pathway ⁿ²		z	p
	M or %	SD	M or %	SD		
7 th Grade (%)	0.28	0.45	0.29	0.45	-0.35	0.724
8 th Grade (%)	0.21	0.41	0.15	0.36	3.26	0.001**
9 th Grade (%)	0.10	0.30	0.17	0.38	-4.67	0.000***
10 th Grade (%)	0.13	0.34	0.15	0.35	-0.89	0.371
11 th Grade (%)	0.21	0.41	0.10	0.31	6.08	0.000***
12 th Grade (%)	0.07	0.26	0.14	0.34	-4.61	0.000***
Male (%)	0.52	0.50	0.48	0.50	1.63	0.104
Hispanic (%)	0.72	0.45	0.66	0.47	2.54	0.011*
White (%)	0.12	0.33	0.12	0.32	0.26	0.792
Asian (%)	0.14	0.34	0.19	0.39	-2.90	0.004**
Black (%)	0.03	0.17	0.03	0.18	-0.62	0.536
Native American (%)	0.05	0.22	0.07	0.25	-1.30	0.193
ELL (%)	0.22	0.41	0.20	0.40	0.88	0.381
RFEP (%)	0.44	0.50	0.43	0.49	0.41	0.685
IFEP (%)	0.03	0.18	0.06	0.24	-2.73	0.006**
EO (%)	0.32	0.46	0.31	0.46	0.03	0.975
FRPL (%)	0.72	0.45	0.66	0.47	2.69	0.007**
AWA Pre	5.11	1.92	5.48	2.38	-3.03	0.002**

N1=40 teachers, 939 students

N2=41 teachers, 887 students

Note: Abbreviations are as follows: ELL (English Language Learner), RFEP (Redesignated as Fluent English Proficient), IFEP (Initially Fluent English Proficient), EO (English Only), FRPL (eligible for free/reduced price lunch)

†p<.10, *p<.05, **p<.01, ***p<.001

Table 3 *Comparison of Pathway and Control Groups on Observation Measures*

	Control ⁿ¹		Pathway ⁿ²			
	M	SD	M	SD	t	p
Teacher's Use of Pathway Strategies						
Year 1 Winter	3.94	1.21	4.39	1.02	-1.67	0.10
Year 1 Spring	4.35	1.38	4.75	0.97	-1.37	0.17
Year 2 Winter	4.32	1.33	4.64	1.10	-1.07	0.29
Year 2 Spring	4.29	1.30	5.19	0.67	-3.66	0.00***
Students' Use of Pathway Strategies						
Year 1 Winter	2.93	1.80	3.34	1.72	-0.89	0.38
Year 1 Spring	3.26	2.07	3.55	1.94	-0.55	0.59
Year 2 Winter	3.96	1.26	4.24	0.99	-0.93	0.36
Year 2 Spring	3.96	1.26	4.59	0.82	-2.21	0.03*

N1=27 teachers

N2=29 teachers

Note: Only teachers who were observed at all four time points were included in these analyses.

*p<.05, **p<.01, ***p<.001

Table 4 *Academic Writing Assessment (AWA) Mean Scores for Year 1*

	Control				Pathway				Difference in gains	<i>d</i>
	n	Pre	Post	Gain	n	Pre	Post	Gain		
All students	851	5.69 (1.92)	5.88 (2.08)	0.19** (1.91)	966	5.72 (2.05)	6.90 (2.18)	1.18*** (1.91)	0.99***	0.52
7 th Grade	164	4.88 (1.75)	5.45 (1.83)	0.56** (1.79)	167	4.46 (1.20)	5.54 (1.85)	1.08*** (1.88)	0.52*	0.30
8 th Grade	170	5.01 (1.54)	5.51 (1.88)	0.50*** (1.82)	211	5.25 (1.80)	6.89 (2.28)	1.64*** (1.84)	1.14***	0.74
9 th Grade	130	5.55 (1.69)	5.36 (1.82)	-0.18 (2.03)	153	5.83 (2.34)	6.75 (2.14)	0.92*** (1.73)	1.11***	0.66
10 th Grade	151	6.36 (1.76)	6.17 (2.00)	-0.19 (1.82)	172	6.08 (1.78)	7.60 (1.87)	1.53*** (1.65)	1.72***	0.98
11 th Grade	128	6.32 (2.42)	6.54 (2.24)	0.22 (1.86)	158	6.57 (2.33)	7.74 (2.21)	1.17*** (2.00)	0.95***	0.39
12 th Grade	108	6.46 (1.60)	6.58 (2.45)	0.12 (2.12)	105	6.65 (1.92)	6.88 (1.82)	0.23 (2.19)	0.11	n.s.
Male	423	5.40 (1.90)	5.54 (2.09)	0.14 (1.95)	448	5.53 (2.12)	6.57 (2.18)	1.05*** (1.83)	0.91***	0.48
Female	428	5.97 (1.89)	6.22 (2.01)	0.25** (1.87)	518	5.89 (1.98)	7.18 (2.14)	1.30*** (1.97)	1.05***	0.56
White	93	5.86 (1.81)	6.25 (1.86)	0.39* (1.86)	90	5.73 (1.86)	6.70 (1.97)	0.97*** (1.72)	0.58*	0.32
Hispanic	573	5.52 (1.95)	5.58 (2.03)	0.06 (1.92)	645	5.34 (1.84)	6.56 (2.10)	1.22*** (1.95)	1.16***	0.59
Asian	148	6.24 (1.82)	6.82 (2.19)	0.59*** (1.88)	178	7.21 (2.30)	8.31 (2.03)	1.10*** (1.82)	0.51*	0.28
Black	24	5.75 (1.59)	5.96 (1.60)	0.21 (1.84)	30	5.33 (1.94)	6.73 (2.42)	1.40** (2.24)	1.19*	0.75
Native Am.	13	5.08 (1.50)	5.38 (1.80)	0.31 (1.97)	22	5.18 (1.18)	6.00 (1.63)	0.82* (1.68)	0.42	n.s.
ELL	159	4.84 (1.50)	4.53 (1.50)	-0.31* (1.71)	168	4.32 (1.23)	5.54 (1.75)	1.22*** (1.63)	1.53***	1.02
RFEP	408	5.74 (1.96)	6.11 (2.09)	0.36*** (1.82)	466	6.02 (2.04)	7.14 (2.08)	1.12*** (1.98)	0.76***	0.39
IFEP	57	6.56 (2.07)	7.19 (2.35)	0.63* (2.01)	88	6.73 (2.35)	7.98 (2.20)	1.25*** (2.04)	0.62 [†]	0.30
EO	227	5.96 (1.86)	6.10 (1.89)	0.13 (2.12)	244	5.75 (1.95)	6.98 (2.21)	1.23*** (1.92)	1.10***	0.59
FRPL eligible	597	5.56 (1.98)	5.68 (2.10)	0.13 (1.91)	681	5.42 (1.90)	6.63 (2.09)	1.21*** (1.91)	1.08***	0.55
FRPL not eligible	254	6.00 (1.71)	6.35 (1.95)	0.35** (1.93)	285	6.43 (2.22)	7.54 (2.26)	1.11*** (1.90)	0.75***	0.44

[†]p<.10, *p<.05, **p<.01, ***p<.001; Standard Deviations in parentheses

Note: The growth of 1 points on the AWA 12-point scale is the equivalent to half a letter grade (from a C to a B-, for example).

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Table 5 *Academic Writing Assessment (AWA) Mean Scores for Year 2*

	Control				Pathway				Difference in gains	<i>d</i>
	n	Pre	Post	Gain	n	Pre	Post	Gain		
All students	616	5.11 (1.92)	5.47 (2.04)	0.36*** (1.93)	634	5.48 (2.38)	7.06 (2.23)	1.58*** (2.00)	1.22***	0.64
7 th Grade	171	4.15 (1.43)	5.18 (1.98)	1.01*** (1.83)	152	4.16 (1.60)	6.77 (1.75)	2.61*** (1.93)	1.61***	1.13
8 th Grade	138	4.64 (1.71)	4.70 (1.54)	0.07 (1.93)	124	5.10 (2.49)	6.42 (2.37)	1.32*** (1.95)	1.26***	0.74
9 th Grade	59	5.08 (1.91)	5.37 (2.12)	0.29 (1.85)	94	5.86 (2.23)	6.64 (1.93)	0.78*** (1.87)	0.49	n.s.
10 th Grade	87	5.03 (1.74)	5.54 (1.86)	0.51* (1.81)	109	5.90 (2.78)	7.37 (2.43)	1.47*** (1.88)	0.96**	0.55
11 th Grade	117	6.35 (1.64)	6.15 (2.18)	-0.20 (2.10)	85	6.42 (2.16)	7.88 (2.25)	1.46*** (1.73)	1.66***	1.01
12 th Grade	44	7.18 (1.98)	7.23 (1.93)	0.05 (1.49)	70	6.71 (1.89)	7.94 (2.30)	1.23*** (2.04)	1.18**	0.60
Male	318	4.73 (1.84)	5.07 (1.89)	0.34** (1.97)	308	5.04 (2.30)	6.70 (2.23)	1.66*** (2.07)	1.32***	0.72
Female	298	5.51 (1.92)	5.89 (2.11)	0.38*** (1.89)	326	5.90 (2.39)	7.40 (2.17)	1.51*** (1.92)	1.13***	0.59
White	69	4.90 (1.70)	5.71 (1.90)	0.80*** (1.90)	66	5.64 (2.23)	7.23 (2.37)	1.59 (2.23)	0.79*	0.46
Hispanic	402	5.03 (1.95)	5.30 (2.09)	0.27** (1.90)	384	4.98 (2.09)	6.72 (2.13)	1.74*** (1.86)	1.47***	0.75
Asian	99	5.90 (1.84)	6.05 (1.98)	0.15 (1.91)	120	7.37 (2.58)	8.49 (1.94)	1.13*** (2.17)	0.98***	0.53
Black	19	4.79 (1.55)	5.68 (2.06)	0.89 (2.40)	18	4.89 (2.00)	6.11 (2.11)	1.22* (1.80)	0.33	n.s.
Native Am.	27	4.15 (1.54)	5.11 (1.45)	0.96* (1.91)	46	4.70 (1.94)	6.30 (1.98)	1.61*** (2.20)	0.65	n.s.
ELL	114	4.23 (1.63)	4.40 (1.52)	0.18 (1.58)	119	4.20 (1.42)	5.55 (1.56)	1.34*** (1.60)	1.16***	0.71
RFEP	305	5.42 (1.87)	5.65 (2.12)	0.23* (1.92)	295	5.62 (2.36)	7.38 (2.12)	1.76*** (2.06)	1.53***	0.82
IFEP	22	5.91 (2.04)	5.64 (1.92)	-0.27 (2.19)	42	7.43 (2.87)	8.43 (2.40)	1.00*** (1.75)	1.27**	0.62
EO	175	5.04 (1.97)	5.82 (2.00)	0.78*** (2.07)	178	5.64 (2.39)	7.24 (2.26)	1.60*** (2.15)	0.82***	0.42
FRPL eligible	471	5.02 (1.94)	5.31 (2.08)	0.29** (1.92)	424	4.92 (2.03)	6.61 (2.08)	1.70*** (1.92)	1.41***	0.73
FRPL not eligible	145	5.40 (1.81)	5.98 (1.82)	0.58*** (1.97)	210	6.62 (2.62)	7.97 (2.24)	1.35*** (2.13)	0.77***	0.43

†p<.10, *p<.05, **p<.01, ***p<.001; Standard Deviations in parentheses

Note: The growth of 1 points on the AWA 12-point scale is the equivalent to half a letter grade (from a C to a B-, for example).

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Table 6 *Effects of Pathway on AWA Gains in Year 1 and Year 2 with Interactions by Race and EL Status*

	Year 1				Year 2			
	(1)	(2)	(3)	(4)	(A)	(B)	(C)	(D)
Pathway	0.99*** (0.22)	0.96*** (0.20)	0.58 (0.31)	1.10*** (0.28)	1.22*** (0.26)	1.30*** (0.20)	0.87* (0.38)	0.85** (0.30)
8 th Grade		0.23 (0.34)	0.22 (0.34)	0.23 (0.34)		-1.10*** (0.32)	-1.10*** (0.31)	-1.10*** (0.31)
9 th Grade		-0.46 (0.36)	-0.44 (0.36)	-0.41 (0.36)		-1.32*** (0.36)	-1.31*** (0.35)	-1.33*** (0.36)
10 th Grade		-0.15 (0.29)	-0.15 (0.30)	-0.12 (0.29)		-0.83* (0.33)	-0.82* (0.32)	-0.85* (0.32)
11 th Grade		-0.15 (0.35)	-0.15 (0.36)	-0.12 (0.35)		-1.21** (0.41)	-1.19** (0.40)	-1.21** (0.40)
12 th Grade		-0.69 (0.41)	-0.69 (0.42)	-0.68 (0.41)		-1.21*** (0.31)	-1.22*** (0.31)	-1.22*** (0.31)
Male		-0.17† (0.10)	-0.17† (0.10)	-0.17† (0.10)		0.05 (0.10)	0.06 (0.10)	0.06 (0.10)
Hispanic		0.03 (0.17)	-0.25 (0.25)	0.04 (0.17)		-0.13 (0.19)	-0.48 (0.29)	-0.15 (0.19)
Asian		0.13 (0.22)	0.18 (0.32)	0.14 (0.21)		-0.50* (0.25)	-0.64 (0.40)	-0.52* (0.25)
Black		0.20 (0.31)	-0.08 (0.42)	0.21 (0.31)		-0.09 (0.38)	0.21 (0.52)	-0.11 (0.39)
Native Am.		-0.31 (0.42)	0.08 (0.57)	-0.30 (0.42)		-0.12 (0.32)	0.33 (0.44)	-0.11 (0.32)
ELL		-0.16 (0.18)	-0.14 (0.18)	-0.32 (0.26)		-0.43* (0.18)	-0.42* (0.18)	-0.60* (0.29)
RFEP		0.11 (0.14)	0.11 (0.14)	0.28 (0.17)		-0.11 (0.18)	-0.11 (0.18)	-0.47† (0.25)
IFEP		0.24 (0.22)	0.29 (0.22)	0.57 (0.30)		-0.59* (0.23)	-0.60* (0.24)	-1.00** (0.34)
FRPL		-0.01 (0.13)	-0.01 (0.13)	-0.02 (0.13)		0.06 (0.16)	0.05 (0.16)	0.05 (0.16)
Pathway *Hispanic			0.56† (0.29)				0.65† (0.37)	
Pathway *Asian			-0.07 (0.40)				0.26 (0.46)	
Pathway *Black			0.58 (0.62)				-0.63 (0.79)	
Pathway *Native Am.			-0.60 (0.76)				-0.73 (0.59)	
Pathway *ELL				0.32 (0.31)				0.34 (0.38)
Pathway *RFEP				-0.32 (0.23)				0.74* (0.33)
Pathway *IFEP				-0.56† (0.39)				0.73 (0.47)
Constant	0.19 (0.15)	0.36 (0.29)	0.54 (0.34)	0.27 (0.32)	0.36 (0.20)	1.44** (0.42)	1.65*** (0.46)	1.68*** (0.44)
N	1817	1817	1817	1817	1250	1250	1250	1250
R ²	0.062	0.090	0.095	0.094	0.089	0.165	0.171	0.171

Standard errors in parentheses. These are adjusted for clustering using the Huber-White “sandwich” estimator in STATA.

†p<.10, *p<.05, **p<.01, ***p<.001

Table 7 *California High School Exit Exam (CAHSEE) Pass Rates in Year 1*

	Control			Pathway			Diff in %
	N	Pass	Percent	N	Pass	Percent	
All 10th grade	262	197	75.19	313	262	83.71	8.52**
Female	128	106	82.81	162	138	85.19	2.37
Male	134	91	67.91	151	124	82.12	14.21**
Hispanic	180	127	70.56	211	168	79.62	9.07*
White	34	31	91.18	29	24	82.76	-8.42
Asian	38	34	89.47	64	62	96.88	7.41
Other	10	5	50.00	9	8	88.89	38.89 [†]
ELL	71	30	42.25	62	32	51.61	9.36
RFEP	105	97	92.38	149	137	91.95	-0.43
IFEP	12	12	100.0	30	28	93.33	-6.66
EO	74	58	78.38	72	65	90.28	11.90 [†]
Non-FRPL	64	54	84.38	106	93	87.74	3.36
FRPL	198	143	72.22	207	169	81.64	9.42*

[†]p<.10, *p<.05, **p<.01, ***p<.001
 Note: Other=Blacks and Native Americans

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Table 8 *California High School Exit Exam (CAHSEE) Pass Rates in Year 2*

	Control			Pathway			Diff in %
	N	Pass	Percent	N	Pass	Percent	
All 10th grade	114	79	69.30	122	107	87.70	18.40***
Female	47	34	72.34	65	59	90.77	18.42**
Male	67	45	67.16	57	48	84.21	17.05*
Hispanic	84	59	70.24	73	62	84.93	14.69*
White	16	11	68.75	15	13	86.67	17.92
Asian	6	5	83.33	28	27	96.43	13.10
Other	8	4	50.00	6	5	83.33	33.33
ELL	30	6	20.00	19	11	57.89	37.89**
RFEP	37	35	94.59	48	45	93.75	-0.84
IFEP	7	6	85.71	16	16	100.0	14.29
EO	40	32	80.00	39	35	89.74	9.74
Non-FRPL	22	17	77.27	48	45	93.75	16.48*
FRPL	92	62	67.39	74	62	83.78	16.39*

†p<.10, *p<.05, **p<.01, ***p<.001

Note: Other=Blacks and Native Americans

Table 9 *Odds ratio coefficients From Logistic Regressions for passing the CAHSEE for Years 1 and 2 with Race and EL Interactions*

	Year 1				Year 2			
	(1)	(2)	(3)	(4)	(A)	(B)	(C)	(D)
Pathway	1.70 (0.91)	1.39 (0.81)	0.44 (-1.11)	2.60 (1.50)	3.16 [†] (1.69)	2.19 (0.96)	1.11 (0.10)	1.47 (0.39)
Male		0.77 (-1.06)	0.77 (-1.03)	0.78 (-0.95)		0.53 [†] (-1.92)	0.53 [†] (-1.83)	0.54 [†] (-1.83)
Hispanic		0.62 (-0.89)	0.36 [†] (-1.64)	0.58 (-0.96)		0.99 (-0.03)	0.84 (-0.48)	0.90 (-0.29)
Asian		1.97 (1.08)	0.81 (-0.50)	2.00 (1.10)		2.02 (0.63)	0.50 (-0.69)	1.62 (0.47)
Other		0.40 (-1.45)	0.12* (-2.28)	0.40 (-1.40)		0.22 [†] (-1.92)	0.14** (-3.15)	0.18* (-2.44)
ELL		0.21*** (-3.22)	0.20** (-3.15)	0.28 [†] (-1.92)		0.07*** (-5.43)	0.06*** (-5.83)	0.04*** (-4.99)
RFEP		2.46** (2.55)	2.37* (2.36)	4.01*** (3.68)		2.19* (2.33)	2.05* (2.18)	3.16* (2.37)
IFEP		3.04 (1.16)	2.77 (1.04)	--		2.63 (0.63)	2.20 (0.53)	--
FRPL		0.78 (-0.77)	0.79 (-0.71)	0.79 (-0.70)		0.64 (-0.51)	0.60 (-0.57)	0.65 (-0.49)
Pathway *Hispanic			3.05 [†] (1.60)	--			1.79 (0.97)	--
Pathway *Asian			6.97* (2.36)	--			12.81 (1.57)	--
Pathway *Other			18.52* (2.37)	--			3.13 (1.17)	--
Pathway *ELL				0.534 (-0.98)				2.88 (1.41)
Pathway *RFEP				0.361 [†] (-1.92)				0.52 (-0.89)
Pathway *IFEP				--				--
Constant	3.03** (2.40)	7.69*** (3.98)	13.50*** (4.20)	6.08*** (3.06)	2.26** (2.37)	10.49*** (3.51)	14.74*** (3.74)	13.26*** (3.16)
Observations	575	575	575	533	236	236	236	220

T-statistics in parentheses. Standard errors have been adjusted for clustering using the Huber-White “sandwich” estimator in STATA. Results for IFEP students have not been reported in models 4 and D due to small sample sizes.

[†]p<.10, *p<.05, **p<.01, ***p<.001

Note: Other=Blacks and Native Americans

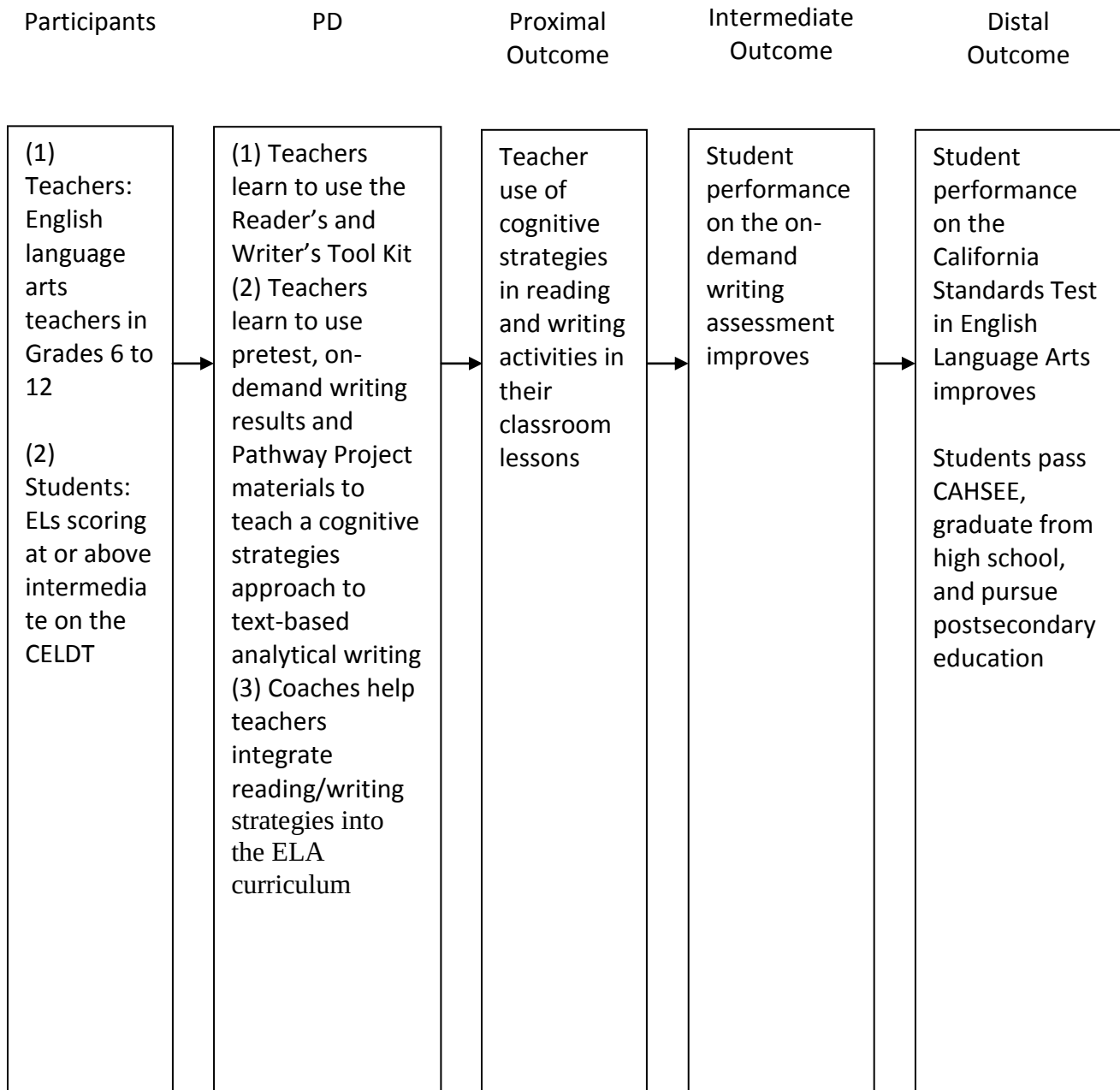


Figure 1. Theory of Change Model for the Pathway Project

Source: Olson, et al., 2012. Reprinted with permission.

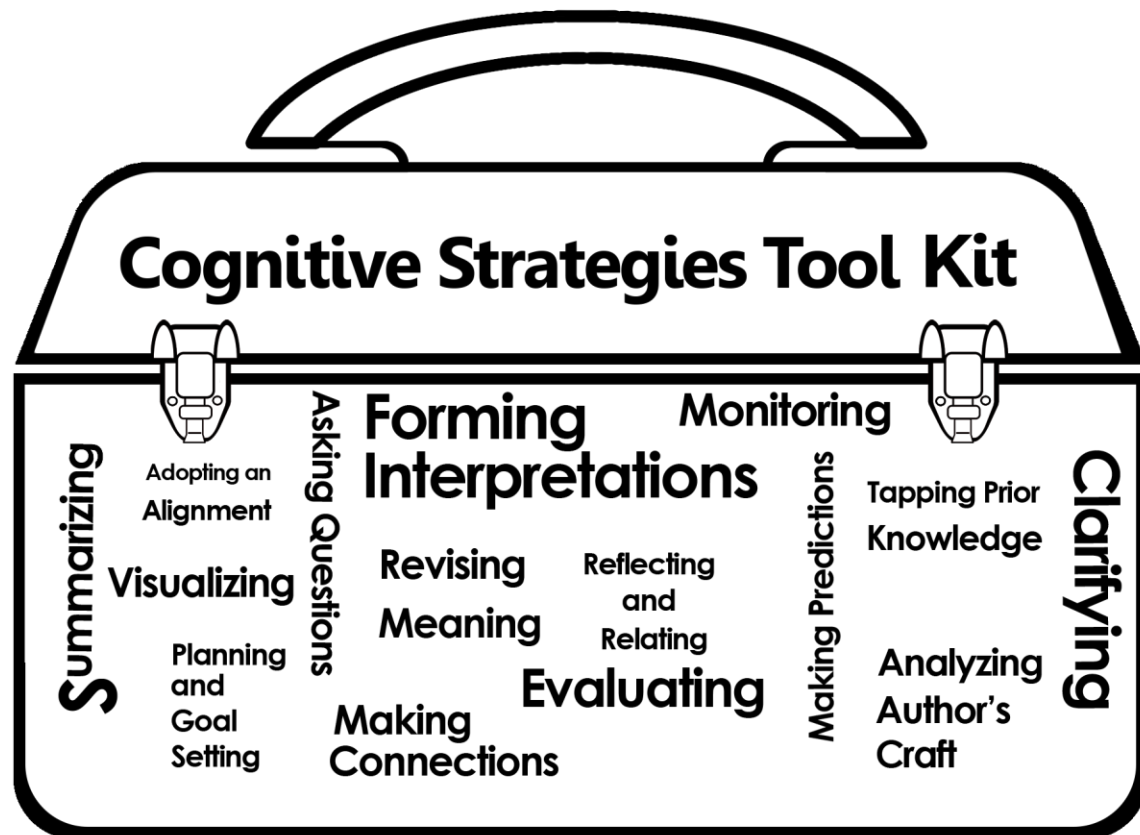


Figure 2. Cognitive Strategies Tool Kit

? How Do I Begin ?

✍ The Introduction to Your Interpretive Essay ✍

4 Parts: HoT S-C T

(HoT S-C Team) = (Hook/TAG/Story-Conflict/Thesis

① Hook: Begin your introductory paragraph with an attention grabber or "hook" to capture the reader's interest. *It might include one of the following:*

- Opening with an exciting moment from the story
- An interesting description
- Dialogue
- Quotation from the text
- A statement to make people think
- An anecdote (a brief story)
- A thought-provoking question (a question that makes people think)

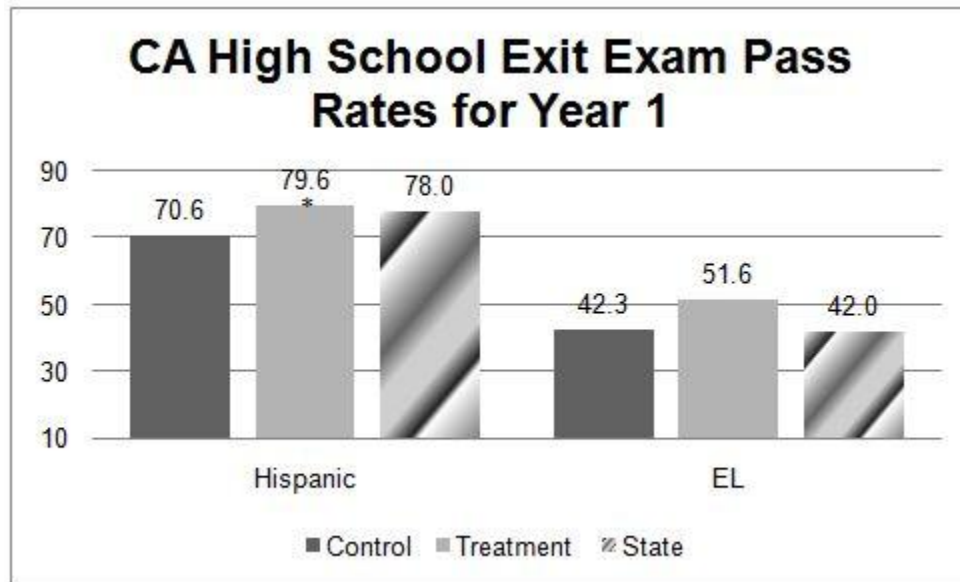
② TAG: Follow the "hook" with a TAG (title/author/genre=type of literature such as short story, narrative, novel, play, poem) that identifies all three parts of TAG for the reader.

③ Summary Statement-Conflict: As a part of the TAG, or right after the TAG, include a brief summary of the story and its conflict. Usually two or three sentences are enough to give background information to the reader about the story and the conflict.

④ Thesis Statement: The thesis statement in an essay is the claim the writer makes in response to the prompt. The thesis statement is the "key" that will "drive" your essay. Do people go on a trip with no idea of where to go? No, they look at a map or check the Internet for driving directions. Your job as a writer is to "map" your essay for the readers. Tell the reader where you will take them.

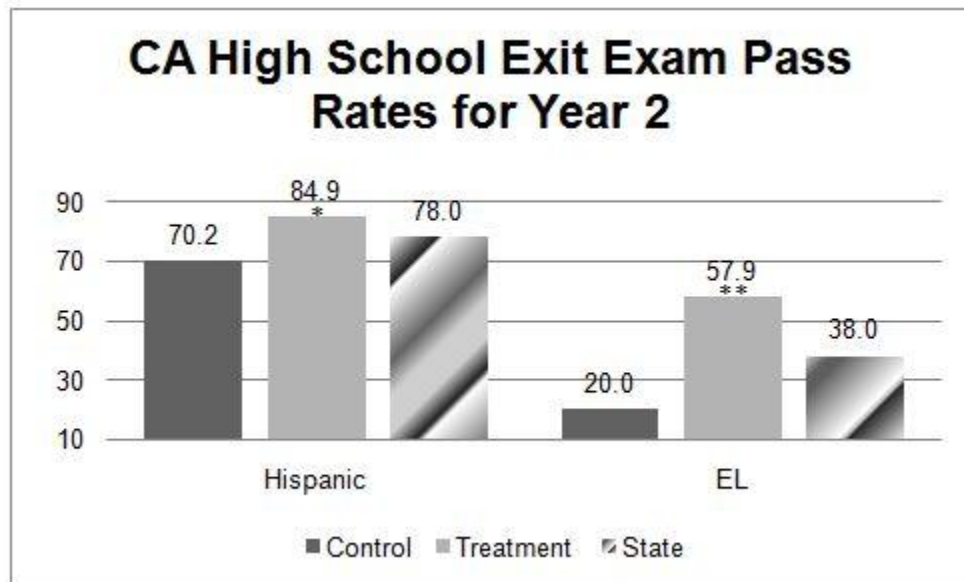
Figure 3. HoT S-C Team

Source: Olson, Scarcella, and Matuchniak, 2013. Reprinted with permission.



†p<.10; *p<.05; **p<.01; ***p<.001

Figure 4. CA High School Exit Exam pass rates for Hispanic and EL students by treatment condition as compared to the state averages (by percentage points) during AY 2012-2013. The star indicates statistical significance in pass rates between Pathway and control groups. Percentages reported come from Table 7.



†p<.10, *p<.05, **p<.01, ***p<.001

Figure 5. CA High School Exit Exam Pass Rates for Hispanic and EL students as compared to the state averages (by percentage points) during AY 2013-2014. Stars indicate statistical significance in pass rates between Pathway and control groups. Percentages reported come from Table 8.

Appendix A

Pre-test written by an 8th grade, Hispanic female, redesignated EL
Received a 5 (3+2) – combined score of two readers
Student Paper
Student Code: 1C124

The theme in the story “The Medicine Bag,” I think that the main theme of this story is tradition. First of all, Martin, the grandson wasn’t as surprise to see grampa coming in his way. He was affraid his friends will juge him since hes grampa didnt look as an indian but was. He has took his friends all about grampa’s stories about animals and people, which made them want to met an indian. As grampa staded in their home or a couple of weeks he dicided to bring his friends over to meet grampa. Martin was happy because they all loved him. Days passed and grampa has told that he has to give him the “Medicine Bag.” He knew that he had to wear it but was also kinda embarresed wearing it to school. But grampa talked to him and gave him the Medicine Bag. That day grampa was sent to the hospital and dieded days later. This made Martin feel sad for dieding, but also was happy to have a part of grampa gaven to him. In conclusion, tradition can bring lots of love and memory to one and traditional ideams is a way to remember a loved one.

Appendix B

Post-test written by an 8th grade, Hispanic female, redesignated EL
Received a 9 (5+4) – combined score of two readers
Student Paper
Student Code: 1C124

He's all there! He's all there!" quotes the narrator as sees his disabled brother smiling at him. In the story, "The Scarlet Ibis," by James Hurst, the narrator learns a valueable lesson to think before you do any action or you might regret it.

In the beginning of the story, the narrator reacts to Doodle, his disabled brother, violently because he wants a brother who's "all there," not a disabled one who's "not there." The narrator has already made plans to kill Doodle by smothering him with a pillow because Doodle "wasn't there." However, his evil idea was washed away when Doodle looked straight at him and smiled. "He's all there, He's all there," yelled the narrator with excitement.

In the middle of the story, the narrator begins to have faith in Doodle finally becoming normal. He is eager to teach Doodle how to walk. "I just can't do it" quotes Doodle, but the narrator doesn't want him to fail. With practice and practice, Doodle finally learned how to walk. As the narrator showed their parents that Doodle could walk, he began to cry. Disappointed and sad, the narrator wanted Doodle to walk only because he was ashamed of having a crippled brother. He says, "They did not know that I did it for myself; that pride, whose slave I was, spoke to me louder than all of their voices."

Towards the end of the story, the family finds a scarlet ibis, weak and sick, dead on the ground. This disappoints Doodle which makes him want to bury the bird. After that, the boys decide to go to the Horsehead Landing where they got to a skiff and floated down the creek with a tide. As darkness descended, they both decided to go home. Tired and frightened, Doodle began to walk closer to the narrator which got him annoyed and he decided to run off without

Doodle. “Brother, Brother, dont leave me! Don’t leave me!” The narrator waited for Doodle to catch up to him but nothing happened. Finally, he went back and found him huddled beneath a red bush. His legs, bent sharply at the knees, had never before seemed so fragile, so thin. The narrator screamed and began to weep. “Doodle!” he cried shaking him; but there was no answer. He layed their crying, sheltering his fallen scarlet ibis.

The Scarlet Ibis is a symbol of the story because Doodle is a weak and sick boy left behind in an unknown place and falls into a bush and dies. The Scarlet Ibis is similar to Doodle; it comes to an unknown place and dies. The theme of this story is to think before you do an action or you might regret it.